DATA ANALYTICS PROJECT DOCUMENTATION

PROJECT TITLE:

Poshan Scheme Coverage and Infrastructure Analysis 2023-2026

PROJECT OVERVIEW:

The ICDS Beneficiary and Infrastructure Dashboard project is designed to analyze and monitor the performance of the Integrated Child Development Services (ICDS) program across different states and districts in India. The dashboard provides insights into beneficiary distribution, worker efficiency, infrastructure availability, Aadhaar verification progress, maternal count, and nutrition-related indicators using interactive visualizations created in Microsoft Excel and Power BI.

The project helps stakeholders understand the current status of ICDS centers, workers, child beneficiaries, infrastructure facilities, and service coverage through data-driven reporting and KPI analysis.

What problem are you solving:

This project solves these issues by creating a centralized and interactive dashboard that allows users to quickly identify:

- States with low Aadhaar coverage
- Areas with infrastructure shortages
- Districts lacking water facilities
- Beneficiary gaps
- Worker shortages
- Nutrition burden trends

The dashboard transforms raw ICDS data into meaningful insights for easier monitoring and decision-making.

Why is this analysis important:

This analysis is important because ICDS plays a critical role in improving child health, nutrition, and maternal care across India. Effective monitoring of the program is necessary to ensure proper resource allocation and service delivery.

The dashboard helps in:

- **Improving Decision-Making**
  Government officials and administrators can quickly identify underperforming regions and take corrective actions.
- **Identifying Infrastructure Gaps**
  The analysis highlights centers without water facilities, own buildings, and sufficient workers, helping authorities prioritize infrastructure development.
- **Monitoring Beneficiary Coverage**

  The dashboard tracks eligible beneficiaries, maternal counts, and child coverage to ensure services reach the intended population.

- **Tracking Aadhaar Verification**

  Aadhaar verification analysis helps monitor digital identity coverage and improve beneficiary authentication processes.

- **Enhancing Worker Efficiency**

  The dashboard measures worker efficiency and workload distribution to support better workforce management.

- **Supporting Data-Driven Governance**

  Interactive visualizations enable faster reporting, transparency, and evidence-based planning for welfare program improvement.

PROJECT OBJECTIVES:

The main objective of this project is to develop an interactive ICDS Beneficiary and Infrastructure Dashboard using Microsoft Excel and Power BI to analyze beneficiary data, monitor infrastructure status, and support effective decision-making in the ICDS program.

- **Analyze Beneficiary Distribution**

  To study the distribution of children, maternal beneficiaries, and adolescent beneficiaries across different states and districts.

- **Monitor ICDS Infrastructure**

  To identify infrastructure-related issues such as:

  Centers without water facilities

  Lack of own buildings

  Worker shortages

  Facility gaps across districts

- **Evaluate Worker Efficiency**
  To measure worker efficiency and analyze workforce availability in ICDS centers.
- **Track Aadhaar Verification Progress**
  To monitor Aadhaar coverage and verification status of beneficiaries across states.
- **Identify Beneficiary Gaps**
  To compare eligible beneficiaries with actual beneficiaries and identify service coverage gaps.
- **Perform State-wise and District-wise Analysis**
  To compare the performance of different states and districts based on:

  Beneficiary count

  Infrastructure availability

  Aadhaar coverage

  Worker performance

DATA SOURCES

- **Dataset Source:** India Data Portal
- **Timeline:** 2023 – 2026
- **Data Size:** 28000+ rows and 20 columns
- **File Format:** CSV File

DATA CLEANING (PREPROCESSING):

The following preprocessing steps were performed to ensure data quality:

- Handled missing values using Average , Averageifs, Median
- Removed duplicate records
- Corrected and changed data types
- Renamed columns for better readability
- Identified and handled outliers

TOOLS USED:

- Excel (Import Data & Impute the Values)
- Excel (Power Query) (For Cleaning)
- Power BI (DAX Functions, Data Modelling, Interactive Dashboard)

DATA TRANSFORMATION:

To enhance analysis, the following transformations were applied:

- Created new calculated columns
- Created calculated measures
- Created Calendar Table
- Derived key performance indicators (KPIs)

DATASET DESCRIPTION:

Table Structure:

Column Name	Data Type
Date	Date
State_Name	Text
State_Code	Number
District_Name	Text
Distrcict_Code	Number
Project	Whole Number
Sector	Whole Number
Anganwadi_Centers	Whole Number
Anganwadi_Workers	Whole Number
Aadhar_Verified_Beneficiaries	Whole Number
Eligible_Beneficiaries	Whole Number
Health_id_Created	Whole Number
AWC Infrastructure Drinking Water Source	Whole Number
AWC Infrastructure Functional Toilets	Whole Number
AWC Infrastructure Own Building	Whole Number
Adolescent_Girls	Whole Number
Children 0-2 Years	Whole Number
Children 3-6 Years	Whole Number
Lacting_Mothers	Whole Number
Pregnant_Womens	Whole Number

Created Columns:

Aadhar Coverage per Centre
Total Childrens
Beneficiaries per Centre
Water Facility
Infrastructure Score
Drinking Water Status
Functional Toilets Status
Buildings Status
Anganwadi Centre Status

Create Calendar Table:

A separate calendar table was created to utilize the date column effectively. This helps in performing time-based analysis such as year-wise trends and comparisons.

Data Modeling:

Table 1	**Relationship**	**Table 2**
Poshan_DB_1	Many-to-One	Calendar Table

DAX Measures Created:

Example Measures:

- Total Centers = SUM(poshan_DB_1[Anganwadi_Centres])
- Workers Shortage = poshan_DB_1[Total Centers]-poshan_DB_1[Total Workers]
- Beneficiary Gap = SUM(poshan_DB_1[Eligible_Beneficiaries]) - SUM(poshan_DB_1[Aadhar_Verified_Beneficiaries])

SCREENSHOTS:

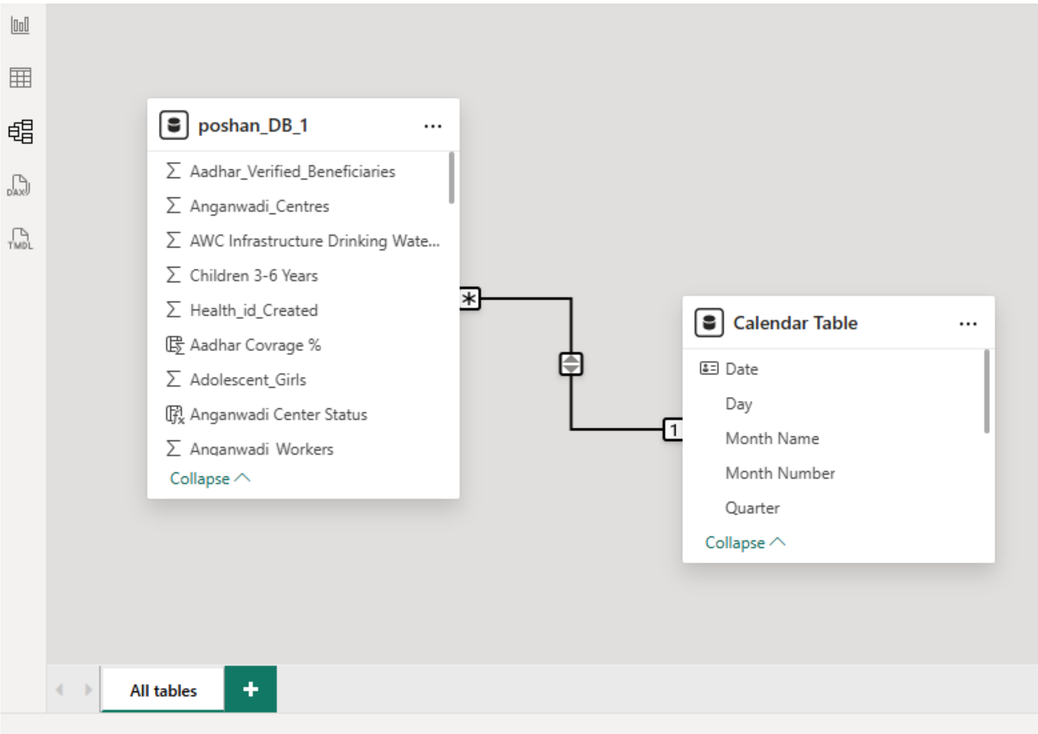
Calendar Table:

```
Calendar Table = ADDCOLUMNS(
    CALENDAR(MIN(poshan_DB_1[Date]),MAX(poshan_DB_1[Date])),
    "Year", YEAR([Date]),
    "Month Number", MONTH([Date]),
    "Month Name", FORMAT([Date], "MMM"),
    "Year-Month", FORMAT([Date], "YYYY-MM"),
    "Quarter", "Q" & FORMAT([Date], "Q"),
    "Day", DAY([Date]),
    "Weekday", FORMAT([Date], "DDD"),
    "Weekday Number", WEEKDAY([Date], 2)
)
```

Date	Year	Month Number	Month Name	Year-Month	Quarter	Day	Weekday	Weekday Number
01-02-2023	2023	2	Feb	2023-02	Q1	1	Wed	3
02-02-2023	2023	2	Feb	2023-02	Q1	2	Thu	4
03-02-2023	2023	2	Feb	2023-02	Q1	3	Fri	5
04-02-2023	2023	2	Feb	2023-02	Q1	4	Sat	6
05-02-2023	2023	2	Feb	2023-02	Q1	5	Sun	7
06-02-2023	2023	2	Feb	2023-02	Q1	6	Mon	1
07-02-2023	2023	2	Feb	2023-02	Q1	7	Tue	2
08-02-2023	2023	2	Feb	2023-02	Q1	8	Wed	3
09-02-2023	2023	2	Feb	2023-02	Q1	9	Thu	4
10-02-2023	2023	2	Feb	2023-02	Q1	10	Fri	5
11-02-2023	2023	2	Feb	2023-02	Q1	11	Sat	6
12-02-2023	2023	2	Feb	2023-02	Q1	12	Sun	7
13-02-2023	2023	2	Feb	2023-02	Q1	13	Mon	1
14-02-2023	2023	2	Feb	2023-02	Q1	14	Tue	2

ble: Calendar Table (1,097 rows) Column: Date (1,097 distinct values)

Data Modelling:



Calculated Columns:

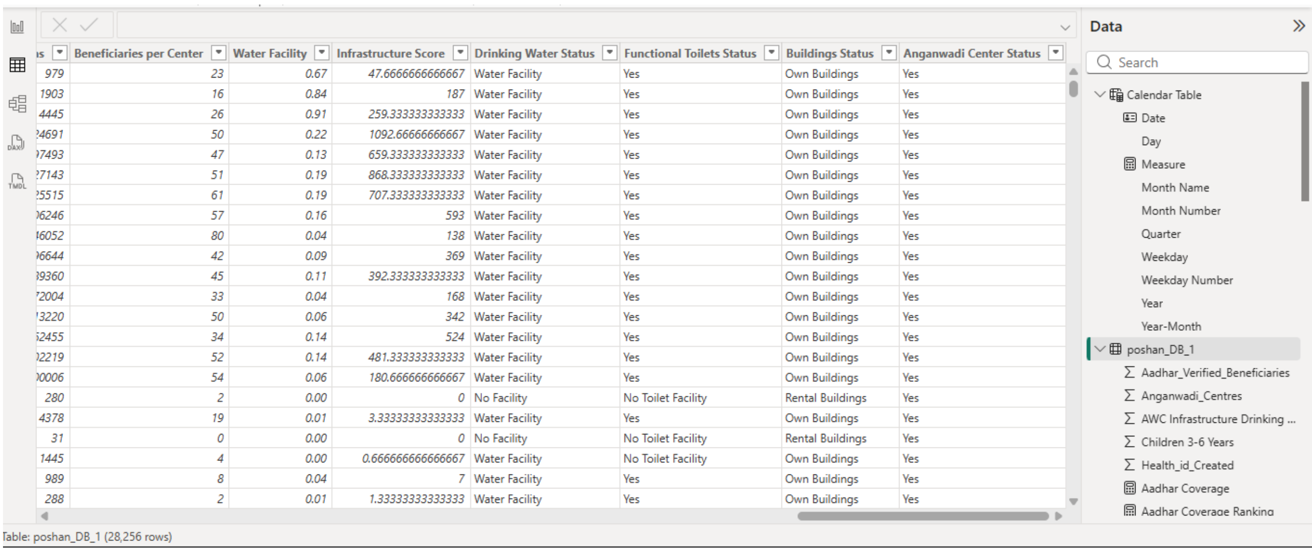
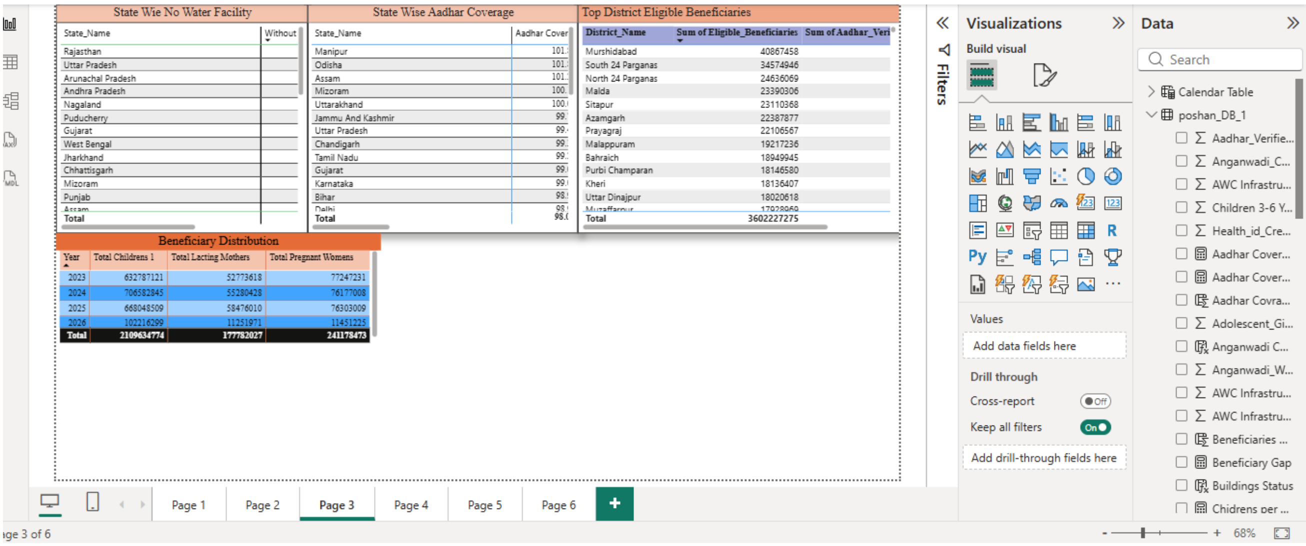
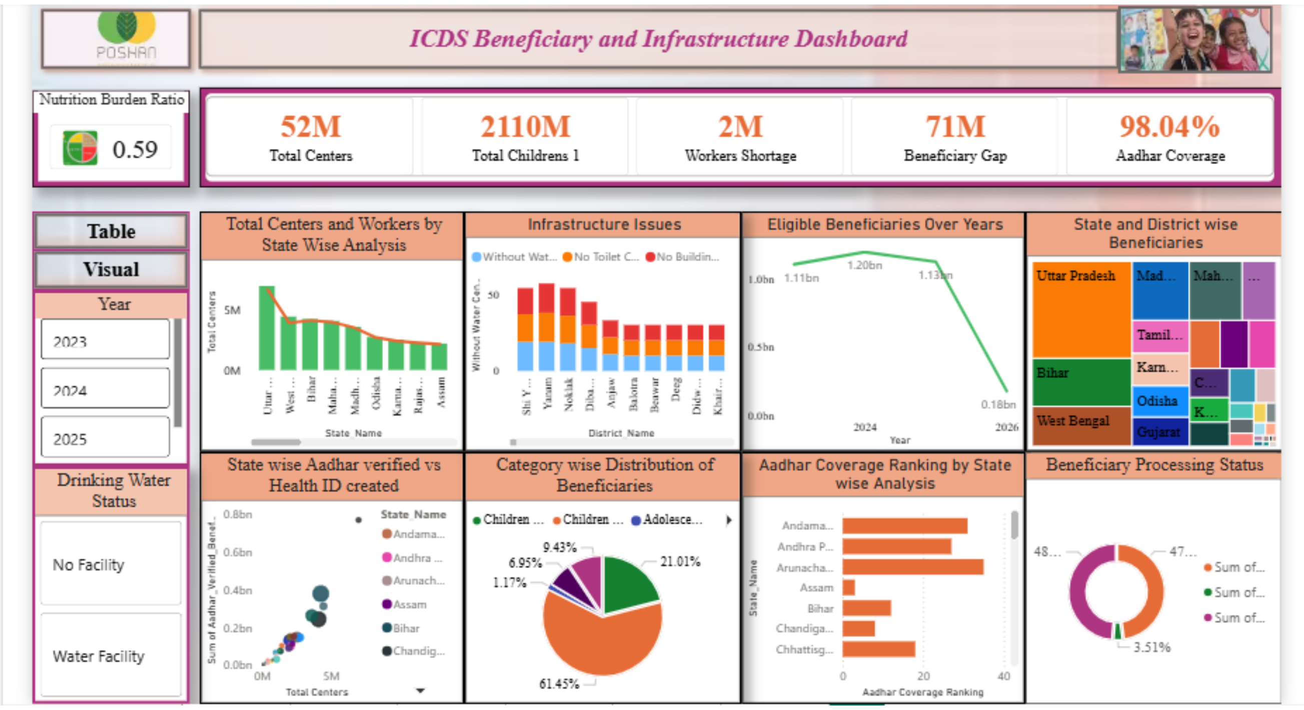


Table & Matrix:



Dashboard:



EXPLORATORY DATA ANALYSIS (EDA) & VISUALIZATION:

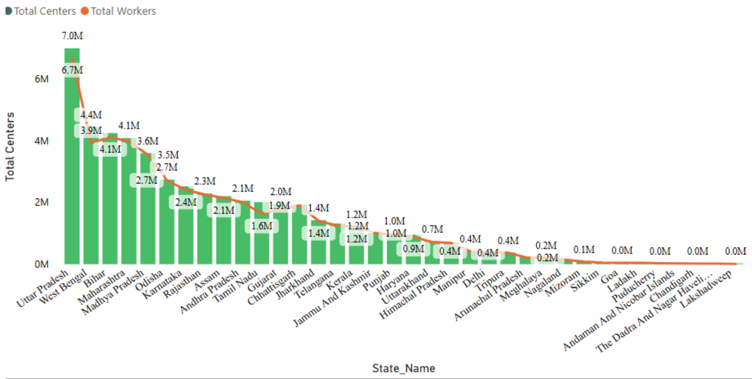
- Missing values and data types were cleaned for accurate analysis.
- Majority of beneficiaries are children compared to women categories.
- Some states have more centers but fewer workers.
- Several centers lack basic facilities like water, toilet, and buildings.
- Year wise analysis shows variation in beneficiary count.

Features Included:

- KPIs
  - Total Centers
  - Total Childrens
  - Workers Shortage
  - Beneficiary Gap
  - Aadhar Coverage
  - Nutrition Burden Ratio
- Slicers
- Drill-down capabilities
- Interactive visuals

INSIGHTS:

❖ State -wise Centers and Workers Distribution



Descriptive Analysis

Uttar Pradesh has the highest number of centres and workers with nearly 6.7M centres and 7.0M workers. West Bengal and Bihar also show high distributions. Smaller states and union territories have comparatively lower numbers.

Diagnostic Analysis

Highly populated states require more centres and workers, which increases the distribution numbers. Rural population and service demand also influence the allocation.

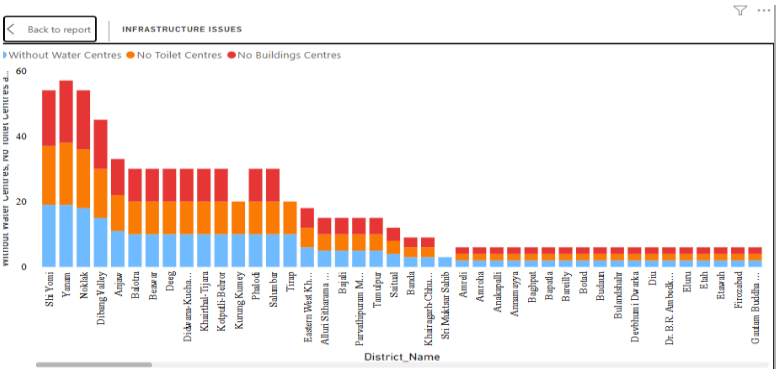
Predictive Analysis

States with larger populations may require more workers and centres in the future to meet growing demand.

Prescriptive Analysis

The government should improve worker allocation and increase facilities in states with high population demand.

❖ Infrastructure Issues:



Descriptive Analysis:

Districts like Sit Yomi (No Water Center 19, No Toilets 18, No Buildings 17), Yaman, Kra Daadi, and Dibang Valley have the highest infrastructure deficiencies, while many other districts show moderate to low issues.

Diagnostic Analysis:

The major reasons may include:

- remote and difficult geographic locations,
- lack of infrastructure investment,
- poor accessibility,
- and limited administrative support.
  High issue counts in northeastern and rural districts suggest development gaps and resource constraints.

Predictive Analysis:

If infrastructure improvements are not prioritized:

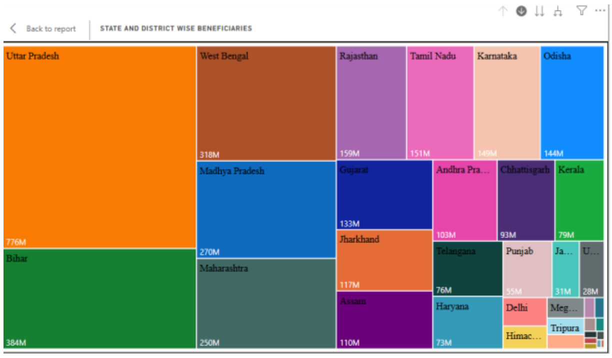
- service quality may decline,
- worker productivity may reduce,
- and operational challenges may increase.

Districts currently facing high deficiencies are likely to continue struggling without targeted intervention.

Prescriptive Analysis

- Prioritize water, toilet, and building facilities in the most affected districts.
- Allocate special infrastructure budgets for remote regions.
- Use phased development plans and regular monitoring.
- Improve coordination with local authorities for faster infrastructure upgrades.

❖ Drill-through Analysis of Beneficiaries(State to District)



Descriptive Analysis:

The tree map shows the distribution of beneficiaries across states and districts in India. Uttar Pradesh has the highest number of beneficiaries (776M), followed by Bihar, West Bengal, Madhya Pradesh, and Maharashtra. Larger states occupy a bigger share of the chart, while smaller states and union territories contribute comparatively fewer beneficiaries.

Diagnostic Analysis:

The higher beneficiary concentration in states like Uttar Pradesh and Bihar is mainly due to large population size, wider rural coverage, and stronger program implementation. States with lower beneficiary counts may have smaller populations, limited outreach, lower awareness, or weaker infrastructure support.

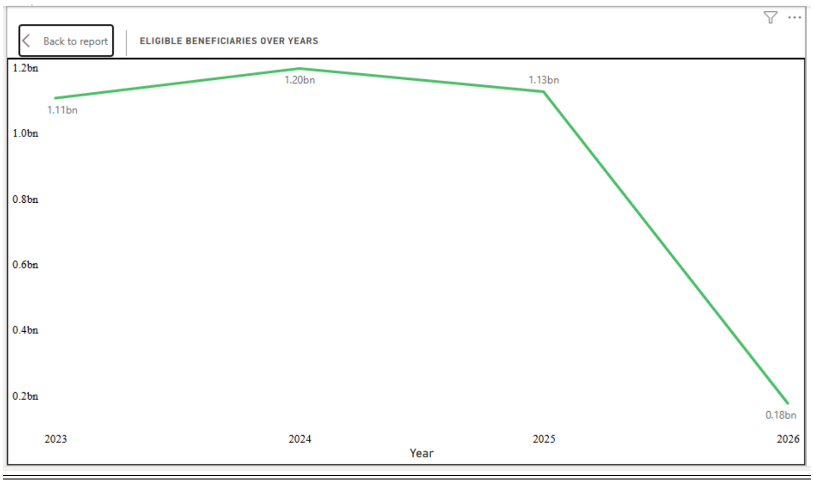
Predictive Analysis:

Based on the current trend, highly populated states are likely to continue leading in beneficiary growth. Emerging states such as Gujarat, Telangana, and Andhra Pradesh may see increasing beneficiary numbers due to better infrastructure development, digital adoption, and expanding service coverage.

Prescriptive Analysis:

To improve overall beneficiary distribution, authorities should strengthen awareness campaigns and outreach programs in low-performing regions, allocate more resources to high-demand states, improve digital enrolment systems, and monitor district-level performance regularly to ensure balanced and efficient program implementation across all regions.

❖ Eligible Beneficiaries Over Years (Line Chart)



Descriptive Analysis

The chart illustrates the trend of eligible beneficiaries from 2023 to 2026. The number of beneficiaries increased from 1.11 billion in 2023 to 1.20 billion in 2024, followed by a slight decrease to 1.13 billion in 2025. A significant decline is observed in 2026, where the beneficiary count dropped sharply to 0.18 billion.

Diagnostic Analysis

The sharp decline in 2026 may have occurred due to several possible factors such as changes in eligibility criteria, incomplete or inaccurate data collection, policy modifications, reduction in beneficiary coverage, or operational/reporting issues. Further investigation is required to identify the exact root cause of the decrease.

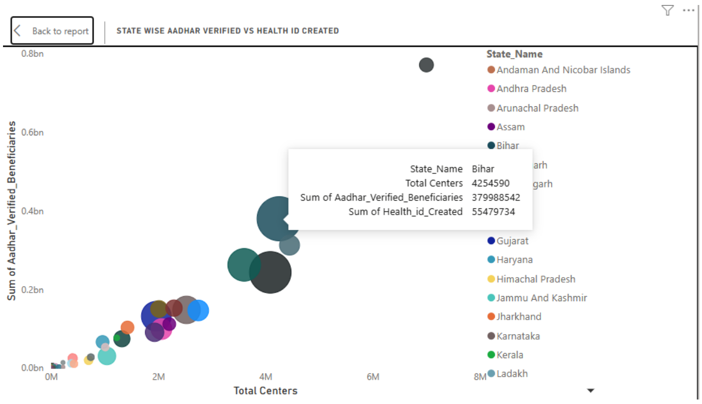
Predictive Analysis

Based on the historical trend, beneficiary counts were relatively stable above 1 billion between 2023 and 2025. If the 2026 decline reflects an actual trend, the number of eligible beneficiaries may continue to decrease in future years. However, if the decline resulted from temporary issues such as incomplete data or reporting errors, the beneficiary count may recover in subsequent years.

Prescriptive Analysis

To address the observed decline, it is recommended to validate the 2026 data for accuracy and completeness. Organizations should review eligibility policies, analyze operational challenges, and improve beneficiary outreach and enrollment processes. Continuous monitoring and early-warning mechanisms should also be implemented to identify unusual variations in beneficiary trends.

❖ State Wise Aadhar verified vs Health ID Created



Descriptive Analysis

The chart presents the relationship between the total number of centers, Aadhaar-verified beneficiaries, and Health IDs created across different states. States with a higher number of centers generally show a higher count of Aadhaar-verified beneficiaries and Health IDs created. Uttar Pradesh stands out with approximately 7 million centers, 771.5 million Aadhaar-verified beneficiaries, and 39.7 million Health IDs created, indicating a significant contribution compared to other states.

Diagnostic Analysis

The variation among states may be influenced by factors such as population size, healthcare infrastructure, digital awareness, government initiatives, and the availability of service centers. States with more operational centers appear to achieve higher Aadhaar verification and Health ID creation rates. Lower-performing states may face challenges such as limited infrastructure, low digital accessibility, or reduced public participation.

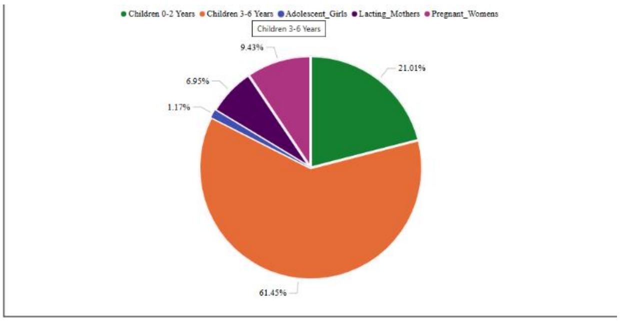
Predictive Analysis

Based on the observed trend, states with increasing numbers of operational centers are likely to achieve higher Aadhaar verification and Health ID creation rates in the future. If current growth patterns continue, large states with strong healthcare infrastructure may continue to dominate beneficiary enrollment and digital health adoption.

Prescriptive Analysis

To improve performance across all states, it is recommended to increase the number of operational centers in low-performing regions, enhance digital health awareness programs, and strengthen infrastructure support. Authorities should also focus on improving accessibility, conducting targeted outreach campaigns, and monitoring state-wise performance regularly to ensure balanced growth in Aadhaar verification and Health ID creation.

❖ Country wise Distribution of Beneficiaries



Descriptive Analysis

The pie chart illustrates the distribution of beneficiaries across different categories. Children aged 3–6 years constitute the largest share with 61.45% of the total beneficiaries. Children aged 0–2 years account for 21.01%, followed by Pregnant Women at 9.43% and Lactating Mothers at 6.95%. Adolescent Girls represent the smallest proportion at 1.17%.

Diagnostic Analysis

The higher percentage of children aged 3–6 years may indicate stronger enrollment and participation in child welfare and nutrition programs targeted toward this age group. The comparatively lower representation of Adolescent Girls could be due to limited awareness, reduced outreach initiatives, or lower participation levels. Variations across categories may also result from demographic distribution, policy focus, and accessibility to welfare services.

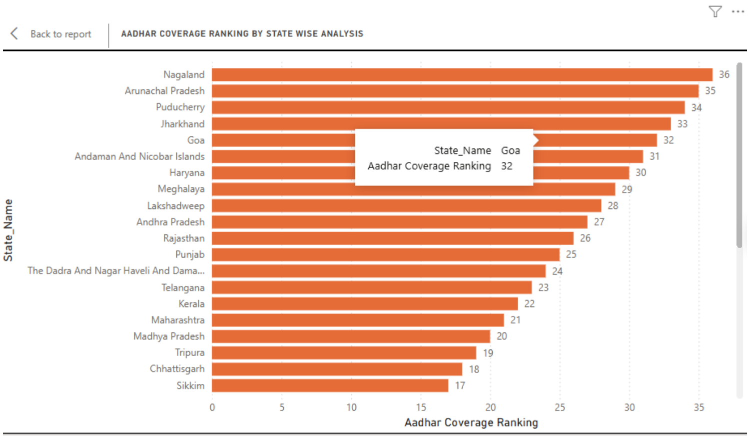
Predictive Analysis

Based on the current distribution, children aged 3–6 years are expected to continue dominating the beneficiary population in future periods. If awareness and outreach programs are improved for underrepresented groups such as Adolescent Girls and Lactating Mothers, their participation rates may increase gradually over time.

Prescriptive Analysis

To ensure balanced beneficiary coverage, authorities should strengthen awareness campaigns and targeted welfare initiatives for underrepresented groups, especially Adolescent Girls and Lactating Mothers. Additional focus should be given to improving accessibility, monitoring category-wise participation, and implementing inclusive policies that encourage equal benefit distribution across all demographic groups.

❖ Aadhar Coverage Ranking by State wise Analysis



Descriptive Analysis

The chart compares Aadhaar coverage rankings across states and union territories. Nagaland (36), Arunachal Pradesh (35), and Puducherry (34) achieved the highest rankings, while Sikkim (17) and Chhattisgarh (18) recorded lower rankings.

Diagnostic Analysis

Higher-ranking states may have better enrollment infrastructure, awareness programs, and administrative monitoring. Lower-ranking states could be facing challenges such as limited accessibility, rural connectivity issues, and lower enrollment efficiency.

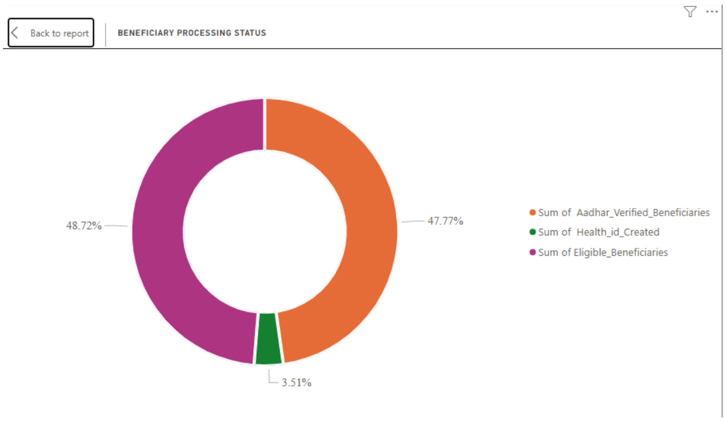
Predictive Analysis

If current trends continue, top-performing states like Nagaland and Arunachal Pradesh are likely to maintain strong Aadhaar coverage, while lower-ranked states may require additional support to improve performance.

Prescriptive Analysis

States with lower rankings should improve Aadhaar enrollment accessibility, strengthen awareness campaigns, and adopt best practices from higher-performing states to increase overall coverage efficiency.

❖ Beneficiary Processing Status



Descriptive Analysis

The chart represents the overall beneficiary processing status by comparing Eligible Beneficiaries, Aadhaar Verified Beneficiaries, and Health ID Created records. Eligible Beneficiaries account for 48.72%, Aadhaar Verified Beneficiaries contribute 47.77%, while Health ID creation remains significantly lower at 3.51%.

Diagnostic Analysis

The analysis indicates that Aadhaar verification coverage is nearly equal to the total eligible beneficiaries, showing strong verification progress. However, the Health ID creation percentage is comparatively very low, which may indicate delays in digital health registration processes, lack of awareness, or incomplete beneficiary onboarding.

Predictive Analysis

If the current processing trend continues, Aadhaar verification is expected to achieve near-complete coverage. However, Health ID creation may continue to lag unless additional digital enrollment initiatives and awareness programs are implemented.

Prescriptive Analysis

To improve beneficiary processing efficiency, focus should be given to increasing Health ID creation through targeted campaigns, field-worker monitoring, digital support systems, and beneficiary awareness programs. Regular tracking of pending registrations can help improve overall processing completion rates.

State wise Ranking for No Water, Building, Toilets Facility

State Wie No Water Facility

State_Name	Without Water Centres	State Rank(No Water Facility)	No Toilet Centres	No Buildings Centres
Rajasthan	87	1	88	87
Uttar Pradesh	77	2	78	80
Arunachal Pradesh	69	3	88	45
Andhra Pradesh	36	4	36	36
Nagaland	27	5	35	27
Puducherry	25	6	25	25
Gujarat	19	7	19	21
West Bengal	16	8	17	16
Jharkhand	14	9	18	16
Chhattisgarh	13	10	13	13
Mizoram	13	10	13	13
Punjab	12	11	10	10
Assam	11	12	11	11
Meghalaya	7	13	7	7
Tamil Nadu	7	13	6	6
Andaman And Nicobar Islands	4	14	4	4
Bihar	4	14	5	7
Haryana	4	14	4	5
Tripura	4	14	4	4
Karnataka	3	15	3	3
Madhya Pradesh	3	15	3	3
The Dadra And Nagar Haveli And Daman And Diu	3	15	3	3
Jammu And Kashmir	2	16	2	8
Odisha	2	16	3	3
Himachal Pradesh	1	17	1	1
Lakshadweep	1	17	1	1
Maharashtra	1	17	1	7
Total	**467**	**1**	**500**	**641**

Descriptive Analysis

The table shows the state-wise number of centres without basic facilities. Rajasthan has the highest number of centres without water facilities (87), followed by Uttar Pradesh (77) and Arunachal Pradesh (69). Rajasthan also records the highest shortage in toilet facilities (88) and building facilities (87). The total number of centres without water facilities is 467, without toilets is 500, and without buildings is 643.

Diagnostic Analysis

The lack of facilities may be due to poor infrastructure, insufficient funding, and remote rural locations. States with difficult geographical conditions show higher shortages.

Predictive Analysis

If improvements are not made, sanitation and public health problems may increase in the future. The affected centres may become less efficient and less accessible.

Prescriptive Analysis

The government should provide better funding, improve water supply systems, construct toilets and buildings, and conduct regular maintenance in highly affected states.

OVERALL KEY INSIGHTS

- Uttar Pradesh has the highest number of centres, workers, and beneficiaries among all states.
- Rajasthan shows the highest infrastructure deficiency in water, toilet, and building facilities.
- Building shortages are higher compared to water and toilet shortages in many districts.
- States with larger populations have more centres, workers, and beneficiaries.
- Remote and rural districts face greater infrastructure challenges.
- Eligible beneficiaries increased from 2023 to 2024 but showed a sharp decline in 2026.
- Unequal infrastructure and resource distribution exist across states and districts.
- Proper planning, funding, and maintenance are necessary to improve facilities and service coverage.

CONCLUSION

The Poshan Statistics analysis highlights the distribution of centres, workers, beneficiaries, and infrastructure facilities across different states and districts in India. The study shows that highly populated states such as Uttar Pradesh have the highest number of centres and beneficiaries, while several districts face shortages in basic facilities like water, toilets, and buildings. The analysis also indicates variations in beneficiary trends over the years. Overall, the project emphasizes the need for better infrastructure development, proper resource allocation, and continuous monitoring to improve the effectiveness of Poshan services and ensure equal access across all regions.

****